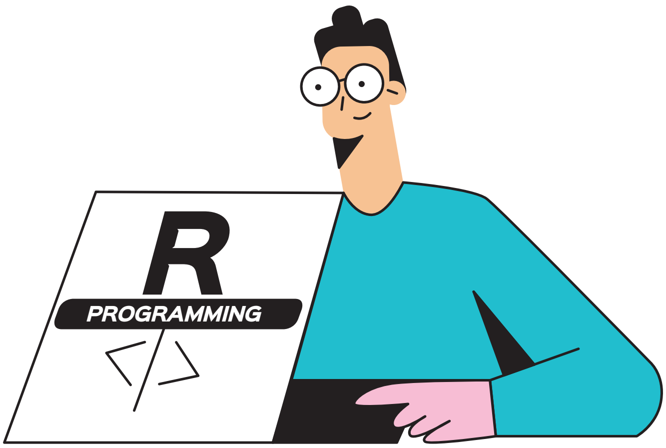
R Series: Probability Distributions

This fifteenth article in the R series will explore more probability distributions and introduce statistical tests.



We will use R version 4.2.1 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version

R version 4.2.1 (2022-06-23) -- "Funny-Looking Kid"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

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You are welcome to redistribute it under the terms of the
GNU General Public License versions 2 or 3.
For more information about these matters see
https://www.gnu.org/licenses/.
```

Welch two sample t-test

Consider the *mtcars* data set in the *lattice* library:

```
> head(mtcars)
            mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4    21.0  6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag 21.0    6  160 110 3.90 2.875 17.02  0  1 4    4
Datsun 710  22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive  21.4  6  258 110 3.08 3.215 19.44  1  0  3  1
Hornet Sportabout 18.7 8 360 175 3.15 3.440 17.02  0  0  3  2
Valiant    18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

The displacement values are shown below:

```
> mtcars$disp
 [1] 160.0 160.0 108.0 258.0 360.0 225.0 360.0 146.7 140.8
167.6 167.6 275.8
[13] 275.8 275.8 472.0 460.0 440.0  78.7  75.7  71.1 120.1
318.0 304.0 350.0
[25] 400.0  79.0 120.3  95.1 351.0 145.0 301.0 121.0
```

We can prepare two data sets, P and Q, from the displacement data, as follows:

```
> P <- mtcars$disp[0:9]
> P
[1] 160.0 160.0 108.0 258.0 360.0 225.0 360.0 146.7 140.8

> Q <- mtcars$disp[10:19]
> Q
[1] 167.6 167.6 275.8 275.8 275.8 472.0 460.0 440.0 78.7 75.7
```

The boxplot for the data sets is shown in Figure 1:

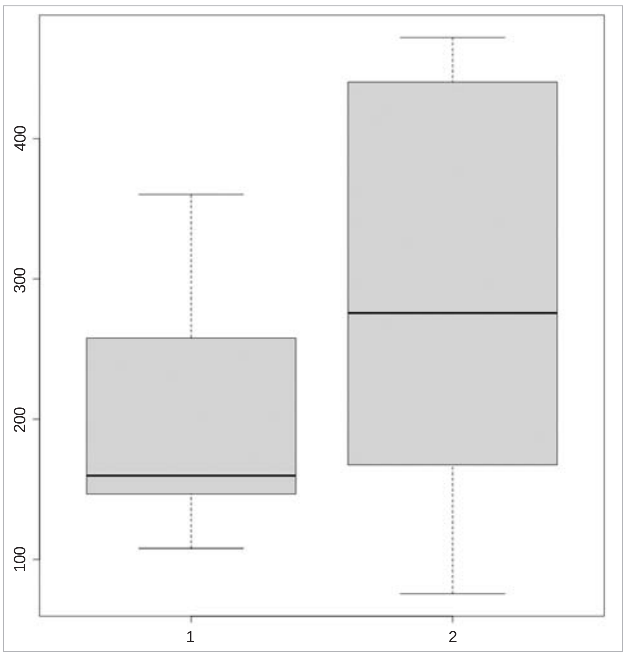

Figure 1: Boxplot *mtcars$disp*